Arduino-Based Power Measurement System

Version 2 Prototype — Code and System Documentation

1. Introduction

Version 2 extends the Arduino-based power measurement prototype by adding a 16 × 2 I²C LCD for local display of measurements. An Arduino Uno R3 interfaces with an INA219 current/voltage sensor and the LCD. The program displays bus voltage and current together, then displays calculated power. This allows readings to be observed without relying on a computer's Serial Monitor.

2. Hardware and Communication

The INA219 measures electrical quantities associated with the load, while the LCD presents selected results. Both modules communicate over I²C. On an Arduino Uno R3, SDA connects to A4 and SCL to A5. The LCD is configured at I²C address 0x27. The INA219 is controlled through the Adafruit_INA219 library, and the LCD through LiquidCrystal_I2C. Both modules can share the I²C data and clock lines when their addresses and wiring are correct.

3. Program Operation

3.1 Libraries and objects

The Wire library provides I²C support. LiquidCrystal_I2C controls the LCD, configured as 16 columns by 2 rows at address 0x27. Adafruit_INA219 provides the functions used to initialise the sensor and obtain readings. The objects `lcd` and `ina` represent the LCD and INA219 modules.

3.2 Initialisation in setup()

The program initialises the LCD and turns on its backlight. It then calls `ina.begin()` to initialise the INA219. If the sensor is not detected, an error message is displayed and the program remains in an infinite loop. If initialisation succeeds, the LCD displays “INA219 connected”.

3.3 Reading measurements

Each loop iteration reads shunt voltage in millivolts, bus voltage in volts, and current in milliamperes. The shunt voltage is stored in `shuntV`, but is not displayed. Power is calculated as `Power = BusV × Current`. Since volts multiplied by milliamperes gives milliwatts, the calculated power is expressed in mW.

3.4 LCD display cycle

The first screen shows bus voltage and current. After a three-second delay, the LCD is cleared and a second screen displays calculated power. This screen remains for another three seconds before the next loop iteration. The final `ina.begin()` call repeats sensor initialisation at the end of each cycle; the sensor-detection check itself is performed in `setup()`.

4. Measurement Quantities

Bus voltage: Voltage measured at the INA219 bus input, in volts (V).
Shunt voltage: Voltage drop across the INA219's current-sense resistor, in millivolts (mV); acquired but not shown on the LCD.
Current: Load current reported by the INA219 library, in milliamperes (mA).
Power: Calculated as bus voltage multiplied by current, in milliwatts (mW).

5. LCD Output Format

The display alternates between a voltage/current screen and a power screen. Numerical values depend on the connected supply and load.

```
Voltage: [value] V
Current: [value] mA
```

```
Power: [value] mW
```

6. Code Listing

```
#include <LiquidCrystal_I2C.h>
#include <Wire.h>
#include <Adafruit_INA219.h>

Adafruit_INA219 ina;
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.init();
  lcd.backlight();
  lcd.setCursor(1, 0);

  if (!ina.begin()) {
    lcd.setCursor(0, 0);
    lcd.print("ERROR: INA219");
    lcd.setCursor(0, 1);
    lcd.print("not found");
    while (1) {
      delay(10);
    }
  }

  lcd.setCursor(0, 0);
  lcd.print("INA219 connected");
}

void loop() {
  float shuntV = ina.getShuntVoltage_mV();
  float BusV = ina.getBusVoltage_V();
  float Current = ina.getCurrent_mA();
  float Power = BusV * Current;

  lcd.setCursor(0, 0);
  lcd.print("Voltage: ");
  lcd.print(BusV);
  lcd.print(" V ");

  lcd.setCursor(0, 1);
  lcd.print("Current: ");
  lcd.print(Current);
  lcd.print(" mA");

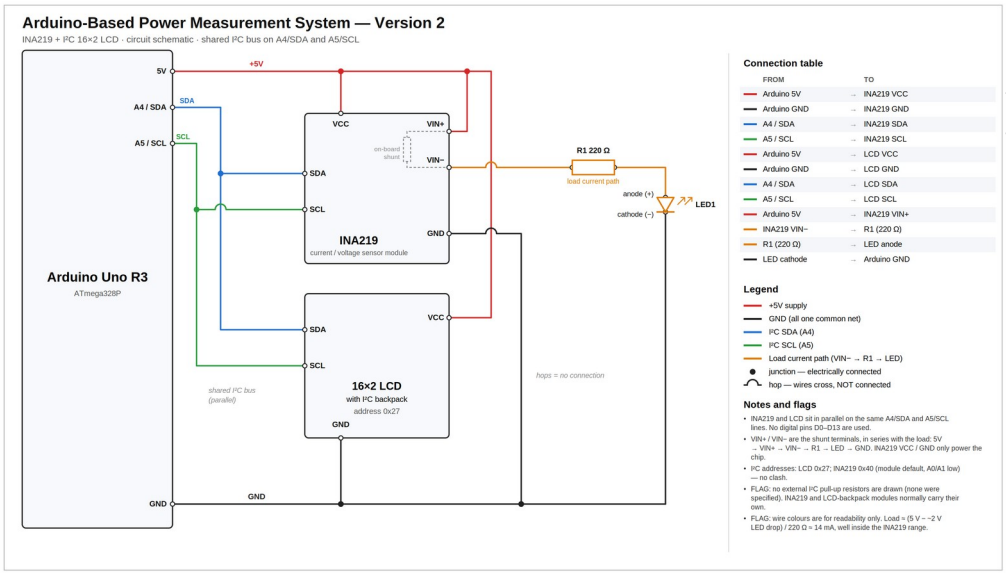
  delay(3000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Power: ");
  lcd.print(Power);
  lcd.print(" mW ");

  delay(3000);
  ina.begin();
}
```

7. Scope of Version 2

Version 2 demonstrates INA219-based measurement acquisition and direct LCD presentation of bus voltage, current, and calculated power for the connected load. The display alternates between two screens at three-second intervals. Shunt voltage is acquired but not presented to the user. The implementation retains a sensor initialisation call at the end of each measurement cycle.

8. Pictures



Circuit Diagram

